

2026년 상반기 연구활동종사자 2차 신규교육

[2026. 6. 9.(화)]

2 소방 안전 (Fire Safety)



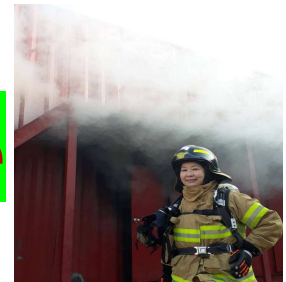
광주과학기술원 안전팀

Fire Safety

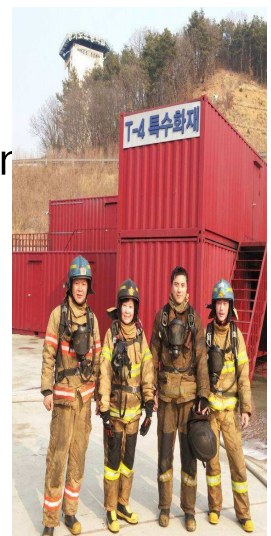
Young hee Kweon



Kweon young hee



- a professor in the Department of Emergency Medical Technology at Daelim University College
- Ph.D. in Engineering, Myongji University
- National First-Class Emergency Medical Technician (EMT-Paramedic)
- 119 EMS Team Leader
- Served 27 years as a firefighter and emergency medical services officer
- Specialist in prehospital emergency care, disaster management, and emergency medical education.



Fire Safety

- 1. Introduction to Fire Safety
- 2. Basics of Fire Science
- 3. Causes of Fire
- 4. Fire Evacuation Procedures
- 5. Classification of Fires
- 6. How to Use a Fire Extinguisher
- 7. How to Use an Indoor Fire Hydrant
- 8. Emergency Escape Device
(Descending Life Line)
- 9. Air Safety Mat (Rescue Air Cushion)

1.fire prevention and safety

- Prevent loss of life and injuries
- Reduce property damage
- Create a safe living environment
- Prevent social and economic losses
- caused by fires

2. Three Elements of Fire (Fire Triangle)

- Heat –

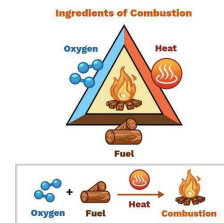
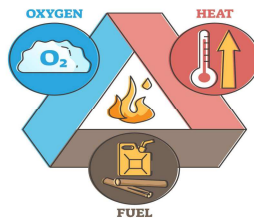
A source of ignition or sufficient temperature to start a fire.

- Fuel –

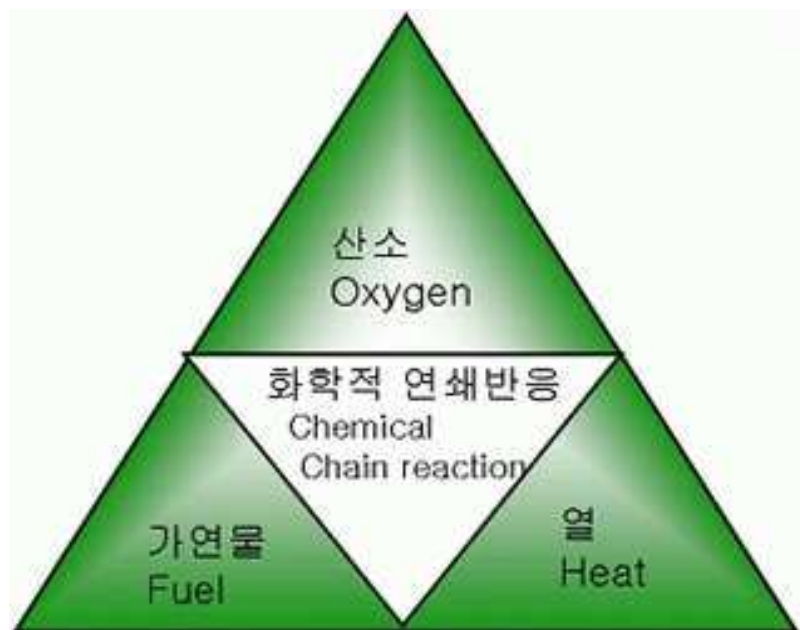
Any combustible material that can burn.

- Oxygen –

An oxidizing agent that supports combustion.



Four Elements of Fire (Fire Tetrahedron)



3. Causes of Fire

- **Electrical Faults**

- **Electrical short circuits, overloaded circuits, and faulty wiring.**

- **Careless Use of Fire**

- **Smoking, unattended cooking, and improper handling of open flames.**

- **Gas Leaks**

- **Leakage of flammable gases and improper use of gas appliances.**

Causes of Fire

- **Heating Equipment Malfunctions**

- **Faulty heaters, boilers, and heating devices.**

- **Flammable Materials**

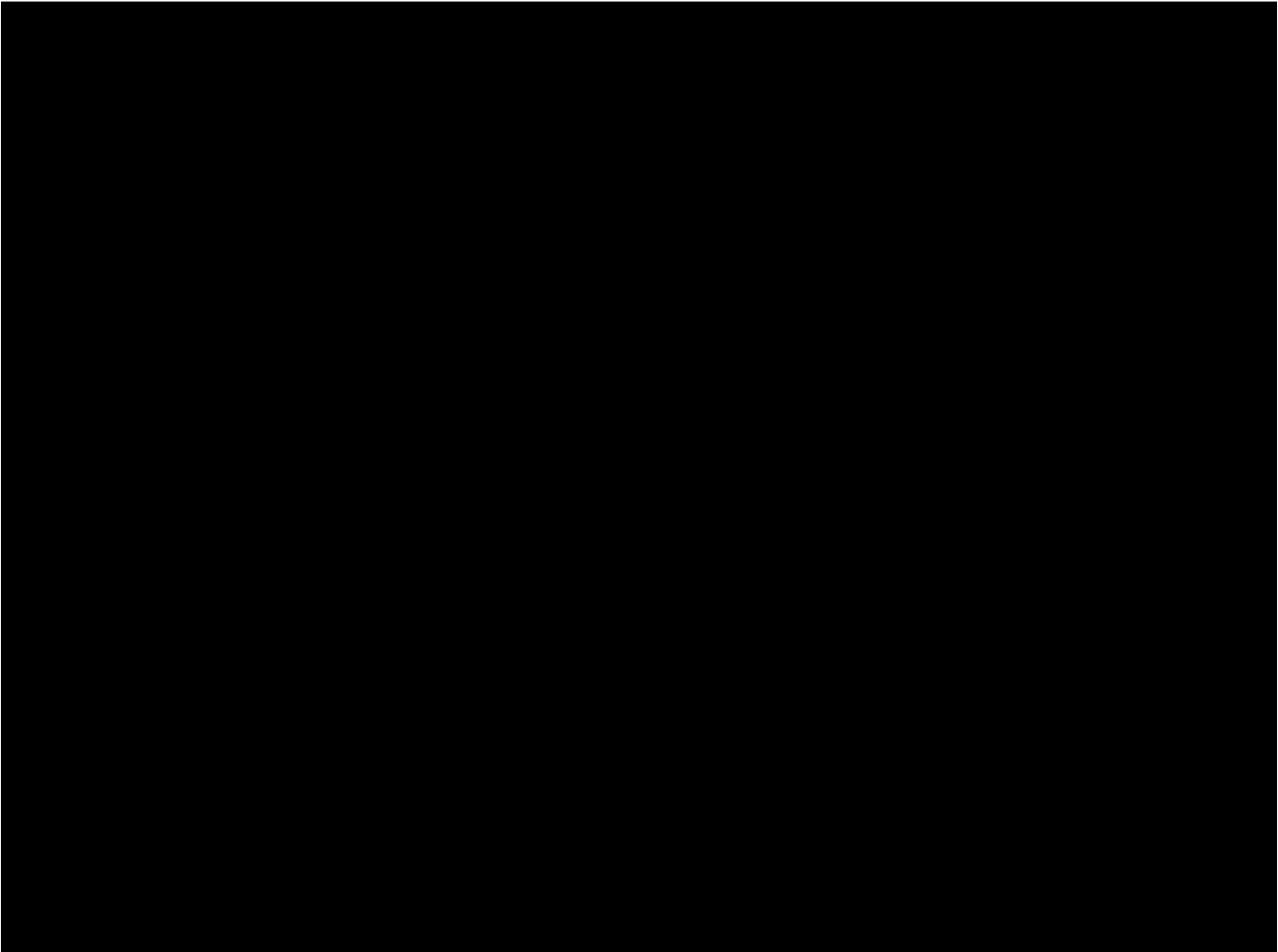
- **Improper storage or handling of combustible and flammable substances.**

- **Mechanical Friction and Overheating**

- **Heat generated by machinery or equipment.**

- **Natural Causes**

- **Lightning strikes and other natural phenomena**



4.Fire Evacuation Procedures

- Stay calm and do not panic.
- Activate the fire alarm and call emergency services.
- Evacuate immediately using the nearest safe exit.
- Do not use elevators; use the stairs instead.
- Stay low to avoid smoke inhalation.
- Cover your nose and mouth with a cloth if possible.
- Close doors behind you to slow the spread of fire and smoke.
- Proceed to the designated assembly point.
- Do not re-enter the building until authorized by emergency personnel.
- Follow the instructions of firefighters and emergency responders.

Fire emergency tip

- In case of the fire emergency , make sure the location and direction of the emergency evacuation.
- Call out loudly, "Fire!" and then press the emergency bell.
- Lower your body as much as possible, quickly evacuate along the direction of guiding light.

[영어]

Tips for fire behavior(화재발생시 행동요령)

1. notify conditions(상 황 전 파)

If a fire occurs, you should quickly notify it around and shout "Fire!" very loud
(화재가 발생하면 신속히 주위에 화재사실을 알려야 한다. "불이야!")



2. Quickly evacuation(신 속 대 피)

Facing fire, you should not turn off the fire alone.
Shall be evacuated outside the building as soon as possible
(불이 나면 절대로 혼자 끄려하여서는 안된다.
건물 외부로 신속히 대피하여야 한다.)

3. reporting 119(119 신고)

You should report "the exact location of where the fire is, and what kind of fire it is" to the fire station.
(정확한 화재발생 위치, 무슨 종류의 화재인지 등 화재정보를 알려 주어야 한다)



4. focusing on early suppression (초기진화주력)

If you find that the fire was not big,
focusing on early suppression using fire extinguisher or fire hydrant.
(화재가 크지 않다고 판단되면 소화기나 옥내소화전을 이용하여 초기진화에 주력한다)

[영어]

Evacuation tips in case of fire (화재발생시 피난요령)

1. In case of fire, what most need to be aware of is suffocating which is caused by poisonous gas and smoke
(화재발생시 가장 주의해야 할 것은 유독가스와 연기로 인한 질식사다)



2. You should be calmly and orderly evacuated according to the guiders who are familiar with the structure
(건물구조에 익숙한 사람의 지시에 따라 침착하고 질서있게 대피하여야 한다)

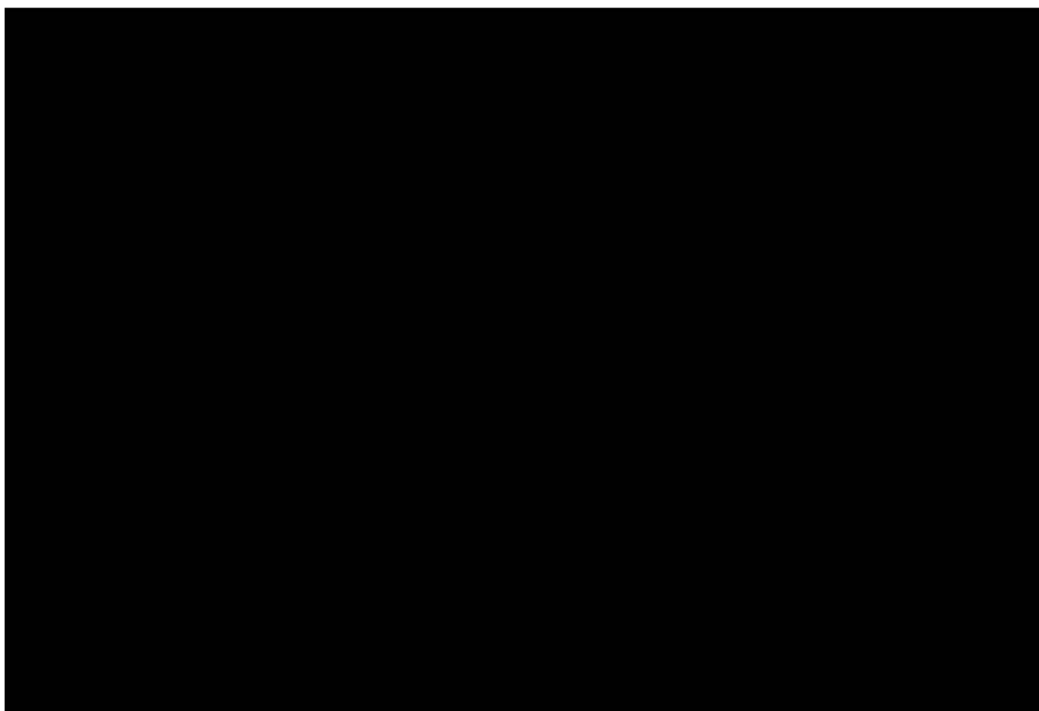
3. Touch the door to make sure if it is hot or not. never open the door when it is hot

(문에 손을 대어 뜨겁지 않은지 확인하고 뜨거울 때에는 절대로 문을 열지 않는다)

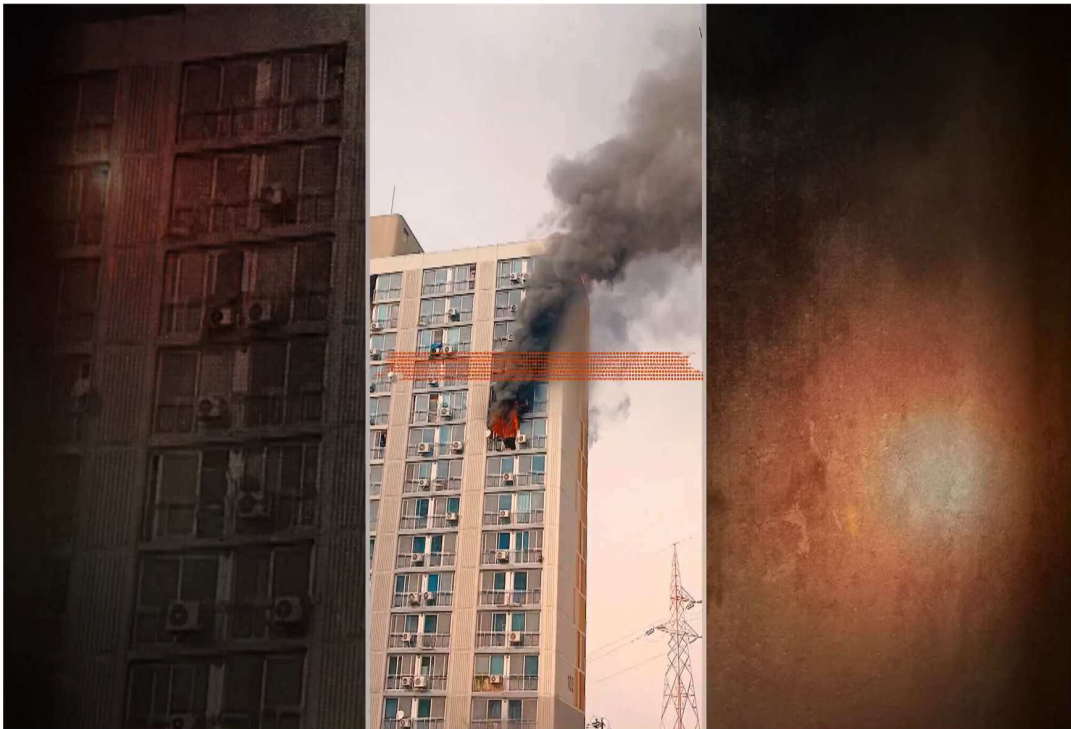


4. while evacuating, stay low and cover mouth with a wet handkerchiefs
(낮은 자세를 취하고 입을 젖은 손수건 등으로 가려서 피난한다.)

Evacuation from a building fire



an apartment fire escape



5. classification of fire

- A grade fire **general fire**

Wood, paper, fiber, synthetic resin, special-purpose, etc.

- B grade fire **Oil Fire**

(Fourth Hazardous Substances)

- C grade fire **electric fire**

(Electrical facilities or equipment in operation)

D grade fire **metallic fire**

Sodium, potassium, aluminum, magnesium, inorganic peroxide, etc.

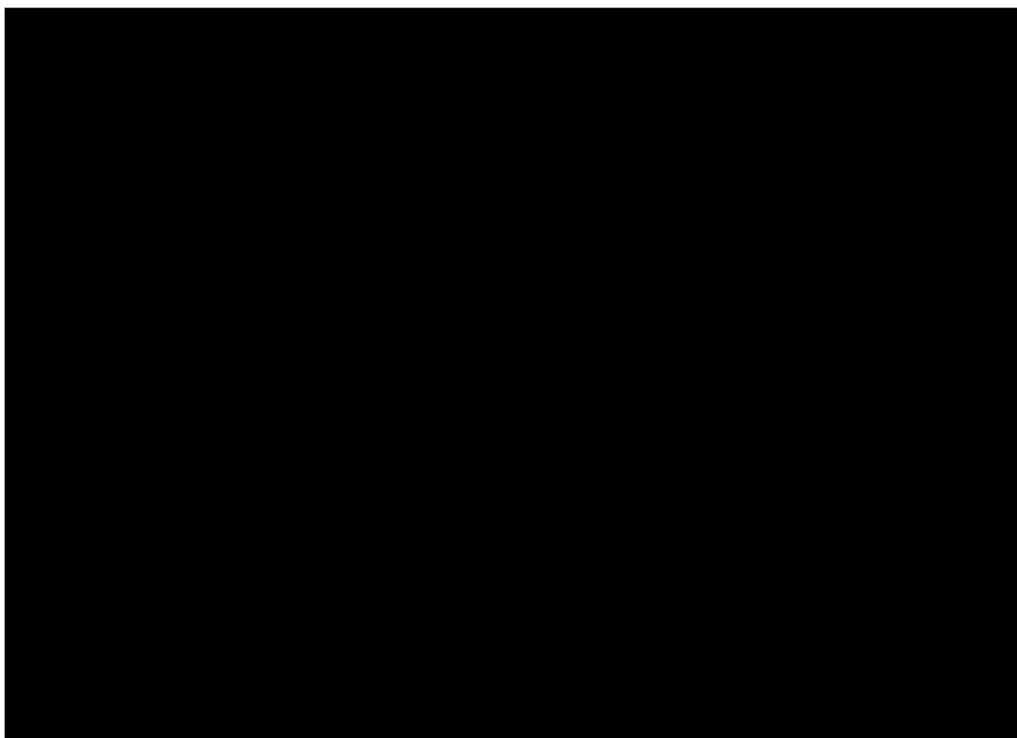
E grade fire **Gas Fire**

LNG, LPG, etc.

K grade fire **kitchen fire**

,vegetable oil, animal fat fire

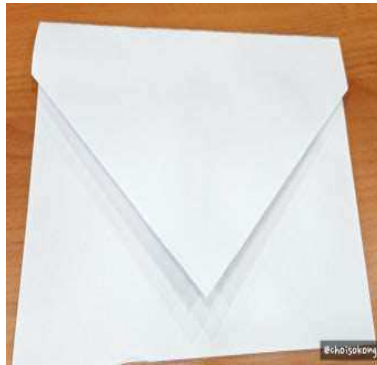
fire extinguisher



classification of fire

- A grade fire general fire

Wood, paper, fiber, synthetic resin, special-purpose etc.



classification of fire

- A grade fire general fire

Wood, paper, fiber, synthetic resin, special-purpose etc.



<http://www.ask-ehs.com>

classification of fire

- A grade fire **general fire** (put out a fire)



classification of fire

- B grade fire Oil Fire , flammable gas fire

(Fourth Hazardous Substances)



classification of fire

- B grade fire Oil Fire , flammable gas fire

In the event of an oil fire,
in order to put out the fire
blocking oxygen
with carbon dioxide, halon, powder
extinguishers, etc
We have to put it down!



classification of fire

– C grade fire **electric fire**

(Electrical facilities or equipment in operation)



classification of fire

– C grade fire **electric fire**

(Electrical car fire)



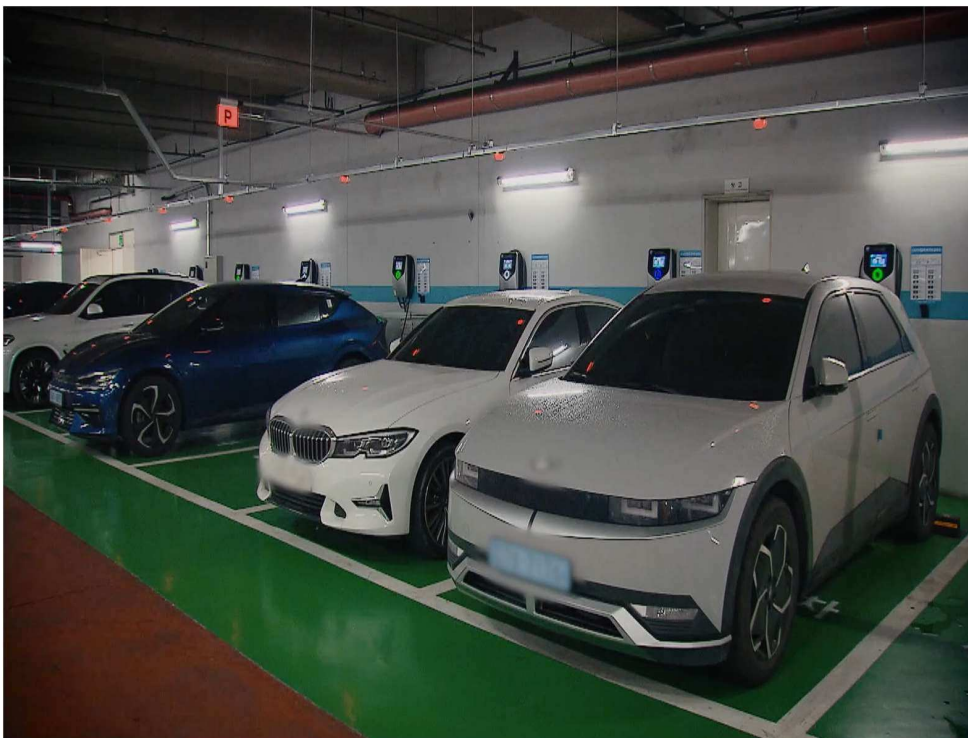
classification of fire

- C grade fire **electric fire**(put out a fire)

Use a foam fire extinguisher or a carbon dioxide fire extinguisher



Evacuation in case of electric vehicle fire



D grade fire **metallic fire**

Sodium, potassium, aluminum, magnesium, inorganic peroxide, etc. lithium(Explosion hazard)



D grade fire **metallic fire**

Sodium, potassium, aluminum, magnesium, inorganic peroxide, etc. lithium(Explosion hazard)



D grade fire **metallic fire** put out a fire)

- dry sand
- d-class fire extinguisher



E grade fire **Gas Fire**

LNG, LPG, etc. factory



K grade fire **kitchen fire**

,vegetable oil, animal fat fire



K grade fire **kitchen fire**

,vegetable oil, animal fat fire



6. How to use fire extinguisher

- Remove the safety pin on the fire extinguisher handle.
- Pick up the handle with one hand, and the hose with the other hand.
- Firmly grab the handle, and inject as like sweeping with broom.

[영어]

fire extinguisher Usage (소화기 사용법)

① Do not panic, remain calm and hold the handle to the access to firearms.
[당황하지 말고 침착하게 손잡이를 잡고 화기 쪽으로 접근]



② Pull the safety pin in front of the handle as hard as
[손잡이 앞쪽에 있는 안전핀을 힘껏 당긴다]



③ Against the wind and grab the handle with a fire hose at
[바람을 등지고 화점을 향해 호스를 빼들고 손잡이를 쥔다]



④ Sprinkle chemicals of fire extinguisher evenly
[골고루 방사한다]



spray fire extinguisher



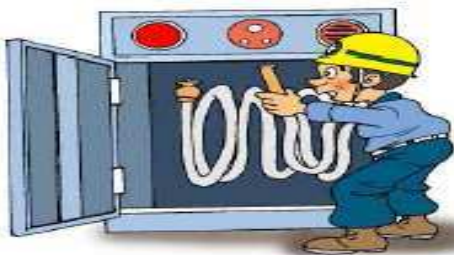
7. How to use the indoor fire hydrant

- Open the fire hydrant and take out the fire hose and nozzle.
- Move to fire place
- Handle opening
- Sprinkle with water

[영어]

Indoor fire hydrant usage(옥내소화전 사용법)

① Open the door and pull out Fire hose and nozzle
(옥내소화전을 열고 소방호스와 노즐을 꺼냄)



② Go to the fire place
(화재발생 장소로 이동)



③ Handle open
(핸들 개방)

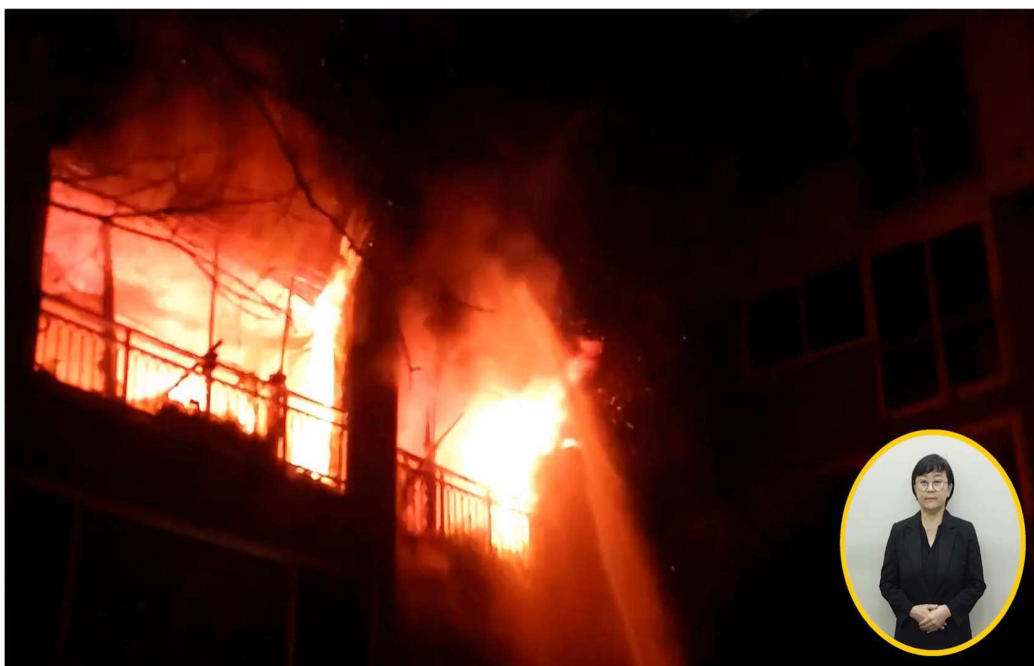


④ Sprinkle water to the fire place
(화점에 방수)





8. a slow-moving escape machine



How to Use an Emergency Descending Device (EDD)

Step 1: Check the Device

Step 2: Put on the Harness

Step 3: Connect to the Rope System

Step 4: Move to the Exit Point

Step 5: Begin Descent

Step 6: Land Safely

9. air safety mat



How to Use a Rescue Air Cushion (Air Mat)

Step 1: Assess the Situation

Step 2: Position the Air Cushion

Step 3: Inflate the Cushion

Step 4: Coordinate with the Victim

Step 5: Perform the Rescue

Step 6: Post-Landing Actions

How to Use a Rescue Air Cushion (Air Mat)

Safety Precautions

- **Only trained personnel should deploy and operate the air cushion.**
- **Never use the cushion without proper inflation.**
- **Keep unauthorized persons away from the landing area.**
- **Follow the manufacturer's operational limits regarding height and capacity.**
- **Maintain constant communication among rescue team members**

Thank you!

